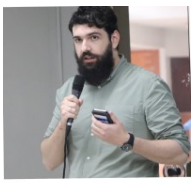
Jhabriel Varela

jvarela@uptp.edu.py
jhabriel
Jhabriel Varela
https://jhabriel.github.io



Employment History

2026 – Present — **Lecturer (Part Time).** *Institute*: National University of Concepción, Concepción, Paraguay.

2025 – Present — **Assistant Professor (Full Time).** Assistant Professor of Applied Mathematics. *Institute:* Polytechnic University Taiwan-Paraguay, Asunción, Paraguay.

2024 – Present — **Lecturer (Part Time).** *Institute*: Polytechnic School, National University of Asunción, San Lorenzo, Paraguay.

2024 – 2025 — **Lecturer (Part Time).** *Institute:* School of Chemical Sciences, National University of Asunción, San Lorenzo, Paraguay.

Lecturer (Part Time). *Institute*: American University, Asunción, Paraguay.

2023 – 2025 — **Researcher (Full Time).** Development of numerical open-source software and *a posteriori* error estimation for mixed-dimensional problems. *Institute*: Polytechnic School, National University of Asunción, San Lorenzo, Paraguay.

2022 – 2023 — **Postdoctoral Fellow in Computational Mathematics for Geophysics (Full Time)** Numerical code verification and software testing. Core developer of PorePy. *Institute:* Center for Coupled Subsurface Processes, University of Bergen, Bergen, Norway.

2019 – 2022 — **VISTA PhD Fellow in Computational and Applied Mathematics (Full Time).** Mathematical modeling of flow processes in fractured porous media. A posteriori error analysis. Numerical code verification. *Institute*: Porous Media Group, University of Bergen, Bergen, Norway.

2018 – 2019 — **Lecturer (Part Time).** *Institute*: Paraguayan-German University, Asunción, Paraguay.

2016 – 2018 — **Numerical Modeling Technician (Part Time).** Modeling and simulation of the Patiño Aquifer as part of the CONACYT project 14-INV-190. *Institute*: Polytechnic School, National University of Asunción, San Lorenzo, Paraguay.

2014 – 2016 — **Research Assistant (Full Time).** Development of numerical code for multiphase flow in porous media. *Institute*: Paraguayan-German University, Asunción, Paraguay.

Education

2019 – 2022 — **Ph.D. in Applied and Computational Mathematics, University of Bergen**
A posteriori error estimation and modeling of unsaturated flow in fractured porous media.
Supervisors: Eirik Keilegavlen, Jan M. Nordbotten and Florin A. Radu.

2016 – 2018 — **M.Sc. in Applied and Computational Mathematics, University of Bergen**
Implementation of an MPFA/MPSA-FV solver for the unsaturated flow in deformable porous media.
Supervisors: Jan M. Nordbotten, Eirik Keilegavlen and Sarah E. Gasda.

2008 – 2014 — **B.Sc. in Chemical Engineering, National University of Asunción**
Modeling and simulation of one-dimensional flow in porous media.
Supervisors: Juan P. Nogues and Christian Schaerer.

Complementary Education

2024 **Diploma in Pedagogical Training in Higher Education**
Community University, Asunción, Paraguay.

2016 **Introduction to Programming with Matlab on Coursera (Online)**
Vanderbilt University, United States of America.
Certificate: `http://coursera.org/verify/C4PXHBHTRV3C`

2014 **ME209x: Thermodynamics on edX (Online)**
Indian Institute of Technology, India.
Certificate: `https://verify.edx.org/cert/4054b339efb44b029de2cdb847b90a40`

CS101.1x Introduction to Computer Programming Part 1 on edX (Online)
Indian Institute of Technology, India. Certificate:
`https://verify.edx.org/cert/c36b3ad6a48b48489cd408165e178810`

Research Publications

Journal Articles

1. I. Stefansson, J. Varela, E. Keilegavlen, and I. Berre, "Flexible and rigorous numerical modelling of multiphysics processes in fractured porous media using PorePy," *Results in Applied Mathematics*, vol. 21, p. 100 428, 2024. DOI: `10.1016/j.rinam.2023.100428`
2. J. Varela, E. Keilegavlen, J. M. Nordbotten, and F. A. Radu, "A model for saturated–unsaturated flow with fractures acting as capillary barriers," *Vadose Zone Journal*, vol. 23, no. 4, e20345, 2024. DOI: `10.1002/vzj2.20345`
3. J. Varela, E. Ahmed, E. Keilegavlen, J. M. Nordbotten, and F. A. Radu, "A posteriori error estimates for hierarchical mixed-dimensional elliptic equations," *Journal of Numerical Mathematics*, vol. 31, no. 4, pp. 247–280, 2023. DOI: `10.1515/jnma-2022-0038`
4. E. Keilegavlen, R. Berge, A. Fumagalli, M. Starnoni, I. Stefansson, J. Varela, and I. Berre, "Porepy: An open-source software for simulation of multiphysics processes in fractured porous media," *Computational Geosciences*, vol. 25, pp. 243–265, 2021. DOI: `10.1007/s10596-020-10002-5`

Preprints

1. J. Varela, C. E. Schaerer, E. Keilegavlen, and I. Berre. "A posteriori error estimates for mixed-dimensional darcy flow using non-matching grids." Submitted on 9 Dec 2025. arXiv: `2512.09087 [math.NA]`.

Conference Proceedings

1. J. Varela and M. A. Dumett, "Functional error estimates for mimetic difference approximations to the poisson problem," in *Anais do CNMAC 2025*, ser. Proceeding Series of the Brazilian Society of Computational and Applied Mathematics, vol. 12, 2026. DOI: `10.5540/03.2026.012.01.0291`
2. J. C. Cabral, G. Espínola, J. Varela, and C. Schaerer, "Adaptive improvements to the lgmres(m, l) method for efficient resolution of sparse linear systems in computational fluid dynamics," in *Proceedings of the XLVI Ibero-Latin American Congress on Computational Methods in Engineering*, Published on 2025-12-01, vol. 5, 2025. URL: `https://publicacoes.softaliza.com.br/cilamce2025/article/view/13651`
3. J. Varela, C. E. Schaerer, and E. Keilegavlen, "A linear potential reconstruction technique based on raviart-thomas basis functions for cell-centered finite volume approximations to the darcy problem," in *Proceeding Series of the Brazilian Society of Computational and Applied Mathematics*, Proceedings of CNMAC 2024, vol. 11, 2025. DOI: `10.5540/03.2025.011.01.0332`

4 S. Gasda, I. Aavatsmark, B. Bohloli, H. Hellevang, J. Nordbotten, M. Wangen, T. Bjørnarå, W. Boon, H. Fazeli, T. Mannseth, M. Nooraiepour, J. Park, E. Skurtveit, S. Tveit, R. van Noort, J. Varela, I. Viken, and S. Thibeau, "Protection of caprock integrity for large-scale CO_2 storage on the Norwegian continental shelf," in *14th Greenhouse Gas Control Technologies Conference (GHGT-14)*, Available at SSRN: `https://ssrn.com/abstract=3365962`, Melbourne, Oct. 21–26, 2018. DOI: `10.2139/ssrn.3365962`

5 L. C. Oporto, D. A. Ramírez, J. Varela, and C. E. Schaerer, "Analysis of contaminant transport under wind conditions on the surface of a shallow lake," in *Mecánica Computacional, Volume XXXIV, Number 31: Fluid Mechanics (C)*, Santa Fe, Argentina: Asociación Argentina de Mecánica Computacional, 2015, pp. 2155–2163. URL: `http://venus.ceride.gov.ar/ojs/index.php/mc/article/view/5135/5067`

Books and Chapters

1 J. Varela, S. E. Gasda, E. Keilegavlen, and J. M. Nordbotten, "A finite-volume-based module for unsaturated poroelasticity," in *Advanced Modeling with the MATLAB Reservoir Simulation Toolbox*, L. Knut-Andreas and O. Møyner, Eds., Cambridge University Press, 2021, ch. 11, pp. 515–548, ISBN: 9781009019781. DOI: `10.1017/9781009019781`

Peer-reviewing Experience

The SMAI Journal of Computational Mathematics

Journal of Hydrology

Mathematics of Computation

SeMA Journal

The Journal of Open Source Software

Engineering Fracture Mechanics

Editorial Experience

2025 – Present **Member of the Editorial Board**. *Journal*: Revista Científica UCOM Scientia. Journal Homepage: https://revista.ucom.edu.py/ucom

Teaching Experience

2026 **Differential and Integral Calculus applied to Optimization Problems**. *Program*: Master's in Engineering and Data Science. *Institute*: National University of Concepción.

Fluid Mechanics. *Programs*: Electromechanical Engineering and Civil Engineering. *Institute*: Polytechnic University Taiwan-Paraguay.

Linear Algebra. *Programs*: Information Engineering and Industrial Engineering. *Institute*: Polytechnic University Taiwan-Paraguay.

Calculus II. *Programs*: Information Engineering and Industrial Engineering. *Institute*: Polytechnic University Taiwan-Paraguay.

2025 **Numerical Methods.** *Programs*: Master's and PhD Program in Computer Science. *Institute*: Polytechnic School, National University of Asunción.

Linear Algebra. *Programs*: Engineering Programs (basic cycle). *Institute*: School of Engineering, National University of Asunción.

Differential Equations. *Programs*: Chemical Engineering and Food Engineering. *Institute*: School of Chemical Sciences, National University of Asunción.

Numerical Calculus. *Program*: Industrial Engineering. *Institute*: Universidad Americana.

Teaching Experience (continued)

Physics I. *Programs*: Electromechanical Engineering and Civil Engineering. *Institute*: Polytechnic University Taiwan-Paraguay.

Calculus I. *Programs*: Electromechanical Engineering, Civil Engineering, Information Engineering, and Industrial Engineering. *Institute*: Polytechnic University Taiwan-Paraguay.

2024 **Flow in Porous Media.** *Program*: Master's and PhD Program in Computer Science. *Institute*: Polytechnic School, National University of Asunción.

Numerical Methods. *Program*: Master's and PhD Program in Computer Science. *Institute*: Polytechnic School, National University of Asunción.

Numerical Calculus. *Program*: Industrial Engineering. *Institute*: American University.

Numerical Calculus. *Programs*: Chemical Engineering and Food Engineering. *Institute*: School of Chemical Sciences, National University of Asunción.

Calculus I. *Programs*: Chemical Engineering and Food Engineering. *Institute*: School of Chemical Sciences, National University of Asunción.

Differential Equations. *Programs*: Chemical Engineering and Food Engineering. *Institute*: School of Chemical Sciences, National University of Asunción.

2018 **Numerical Methods for Engineering.** *Program*: Industrial Engineering. *Institute*: School of Engineering Sciences, Paraguayan-German University.

Ordinary Differential Equations. *Program*: Industrial Engineering. *Institute*: School of Engineering Sciences, Paraguayan-German University.

Introduction to Programming with MATLAB. *Program*: Industrial Engineering. *Institute*: School of Engineering Sciences, Paraguayan-German University.

Supervision

2025 **Gustavo Espínola**
Co-Supervisor. M.Sc. in Computer Science. National University of Asunción, Paraguay.
Thesis: A Numerical Investigation on Preconditioners and Iterative Methods for Solving the Poisson Equation with High-Order Mimetic Operators.

2020 **Victoria Nymo**
Co-Supervisor. M.Sc. in Applied and Computational Mathematics. University of Bergen, Norway.
Thesis: Mathematical and Numerical Methods for physical regulations of medical image.

2019 **Renato González**
Main Supervisor. B.Sc. in Business Engineering. Paraguayan-German University, Paraguay.
Thesis: Análisis Comparativo del Mercado Farmaceútico y su Impacto Económico España y Argentina.

2018 **Sebastián Zaldívar**
Main Supervisor. 2018. B.Sc. in Business Engineering. Paraguayan-German University, Paraguay.
Thesis: Aplicación de la Aplicación de la Logística Inversa en la Región Latinoamericana.

Skills

Languages	Spanish, English, Portuguese.
Coding	Python, MATLAB, LaTeX, C++.
Misc.	Academic research, teaching, training, consultation, LaTeX typesetting and publishing.

References

Christian Schaerer, D.Sc. *email:* cschaer@pol.una.py

Joel Prieto, Ph.D. *email:* jprieto@uptp.edu.py